

ADT ARRAY

QUESTIONS 10 / 10 completed

Integer Array

INTEGER ARRAY

Sample Input:
Enter number of elements: 4
Enter elements:
10 20 30 40
Sample Output:
Using Array Indexing:
arr[0] = 10
arr[1] = 20
arr[2] = 30
arr[3] = 40

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i;
5     int arr[100];
6     int *ptr;
7     printf("Enter number of elements: ");
8     if (scanf("%d", &n) != 1) return 1;
9     printf("Enter elements:\n");
10    for (i = 0; i < n; i++) {
11        scanf("%d", &arr[i]);
12    }
13    ptr = arr;
14    printf("Using Array Indexing:\n");
15    for (i = 0; i < n; i++) {
16        printf("arr[%d] = %d\n", i, arr[i]);
17    }
18    printf("\nUsing Pointer Notation:\n");
19    for (i = 0; i < n; i++) {
20        printf("*(ptr+%d) = %d\n", i, *(ptr + i));
21    }
22 }
```

OUTPUT

INPUT

Your I

QUESTIONS 10 / 10 completed

Factorial

FACTORIAL

"Write a C program to find the factorial of a given number using recursion.
Sample Input: Enter a number- 6
Sample Output: Factorial of 6 is 720
"

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```
1 #include <stdio.h>
2 long long factorial(int n) {
3     if (n == 0 || n == 1) {
4         return 1;
5     }
6     return n * factorial(n - 1);
7 }
8
9 int main() {
10    int num;
11
12    printf("Enter a number- ");
13    if (scanf("%d", &num) != 1) {
14        printf("Invalid input.\n");
15        return 1;
16    }
17
18    if (num < 0) {
19        printf("Factorial is not defined for negative numbers.\n");
20    } else {
21        long long result = factorial(num);
22    }
23 }
```

INPUT

Your I

QUESTIONS 10 / 10 completed

Array Rotation

ARRAY ROTATION

"Implement array rotation (left/right) for arrays of size 10 by K positions without using extra space. Use a reversal algorithm. The program must validate K ($0 \leq K < N$), show original→rotated→reversal steps visually, compute the sum

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```
1 #include <stdio.h>
2 void reverse(int arr[], int start, int end) {
3     while (start < end) {
4         int temp = arr[start];
5         arr[start] = arr[end];
6         arr[end] = temp;
7         start++;
8         end--;
9     }
10 }
11
12 void printArray(int arr[], int n) {
13     printf("[");
14     for (int i = 0; i < n; i++) {
15         printf("%d", arr[i]);
16         if (i < n - 1) printf(" ");
17     }
18     printf("]\n");
19 }
20
21 int main() {
```

INPUT

Your INPUT g

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Sparse Matrices

SPARSE MATRICES

Represent two sparse matrices (mostly zeros) using arrays of triples [row, col, value]. Read dimensions and non-zero terms for two matrices. Add them, store result in sparse form (sorted by row then col), and display dense matrix form. Max 10x10;

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```
1 #include <stdio.h>
2
3 // Structure to represent a non-zero element as a triple
4 typedef struct {
5     int row;
6     int col;
7     int val;
8 } Triple;
9
10 int main() {
11     int r1, c1, n1, r2, c2, n2;
12     Triple m1[100], m2[100], sum[200];
13     int sum_count = 0;
14
15     // Input for Matrix 1
16     if (scanf("%d %d %d", &r1, &c1, &n1) != 3) return 0;
17     for (int i = 0; i < n1; i++) {
18         scanf("%d %d %d", &m1[i].row, &m1[i].col, &m1[i].val);
19     }
20
21     // Input for Matrix 2
```

INPUT

Your INPUT goes here!

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QUESTIONS 10 / 10 completed

Sparse Matrices

SPARSE MATRICES

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C

Run

Save

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12     Triple m1[100], m2[100], sum[200];
13     int sum_count = 0;
14
15     // Input for Matrix 1
16     if (scanf("%d %d %d", &r1, &c1, &n1) != 3) return 0;
17     for (int i = 0; i < n1; i++) {
18         scanf("%d %d %d", &m1[i].row, &m1[i].col, &m1[i].val);
19     }
20
21     // Input for Matrix 2
```

INPUT

Your INPUT goes here!

QUESTIONS 10 / 10 completed

Array

ARRAY

Write a C program to find the 5th element in the given array/
Sample Input: a=[10,20,30,40,50,60]
Sample output: 50

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C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int a[] = {10, 20, 30, 40, 50, 60};
5
6     int n = sizeof(a) / sizeof(a[0]);
7
8     if (n >= 5) {
9         printf("%d", a[4]);
10    } else {
11        printf("Array is too small.");
12    }
13
14    return 0;
15 }
```

INPUT

Your INPUT goes here!

QUESTIONS 10 / 10 completed

Fibonacci series (N

FIBONACCI SERIES (N ≤ 50)

Develop a C program to print the first N terms of the Fibonacci series (N ≤ 50) without using recursion. Use two variables to generate the series iteratively and store the series in an array. The program should also calculate and display the sum of

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Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i;
5     long long fib[50];
6     long long sum = 0;
7     long long max_term = 0;
8
9     printf("Enter number of terms (1-50): ");
10    if (scanf("%d", &n) != 1 || n < 1 || n > 50) {
11        printf("Invalid input! Please enter terms between 1 and 50.\n");
12        return 0;
13    }
14
15    for (i = 0; i < n; i++) {
16        if (i == 0) {
17            fib[i] = 0;
18        } else if (i == 1) {
19            fib[i] = 1;
20        } else {
21            fib[i] = fib[i - 1] + fib[i - 2];
```

OUTPUT

QUESTIONS 10 / 10 completed

Unsorted Array

UNSORTED ARRAY

Write a program to find the missing element for the given unsorted array of sequence numbers.
Sample Input: a=[10,9,7,4,3,1,2]
Sample output: 5,6,8

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 void findMissingElements(int arr[], int n) {
5     if (n == 0) return;
6
7     int min = arr[0];
8     int max = arr[0];
9
10    for (int i = 1; i < n; i++) {
11        if (arr[i] < min) min = arr[i];
12        if (arr[i] > max) max = arr[i];
13    }
14
15    int range = max - min + 1;
16    int *present = (int *)calloc(range, sizeof(int));
17
18    for (int i = 0; i < n; i++) {
19        present[arr[i] - min] = 1;
20    }
21 }
```

INPUT

Your INPL

QUESTIONS 10 / 10 completed

Fib series-iterative ▾

FIB SERIES-ITERATIVE & RECUR

Write a C program to generate Fibonacci series up to N terms using both iterative and recursive methods. The program should display both outputs and compare execution time in milliseconds. Also display total recursive calls made.

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C ▾

Run Save

```
1 #include <stdio.h>
2 #include <time.h>
3 int recursive_calls = 0;
4
5 int fibonacci_recursive(int n) {
6     recursive_calls++;
7     if (n <= 1) {
8         return n;
9     }
10    return fibonacci_recursive(n - 1) + fibonacci_recursive(n - 2);
11 }
12
13 void print_fibonacci_iterative(int n) {
14     int t1 = 0, t2 = 1, nextTerm;
15     for (int i = 1; i <= n; ++i) {
16         printf("%d ", t1);
17         nextTerm = t1 + t2;
18         t1 = t2;
19         t2 = nextTerm;
20     }
21     printf("\n");
22 }
```

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Structure ▾

STRUCTURE

Write a C program to store student details (Roll No, Name, Marks) using structure and save them into a file. Display highest mark, class average, and total records stored.

Sample Input:
Enter number of students 2

PROGRAM EDITOR

C ▾

Run Save

INPUT

Your INPUT goes here!

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4
5 struct Student {
6     int rollNo;
7     char name[50];
8     float marks;
9 };
10
11 int main() {
12     int n, i;
13     float sum = 0, maxMarks = -1.0;
14     int maxIndex = 0;
15
16     printf("Enter number of students: ");
17     if (scanf("%d", &n) != 1 || n <= 0) {
18         printf("Invalid number of students.\n");
19         return 1;
20     }
21 }
```